YOURWISH

11. COMPLICATIONS OF ASOM AND CSOM

ROUTES OF SPREAD OF INFECTION FROM MIDDLE EAR

00:00.34

- **Natural Communications**
  - Oval window and Round window
  - These are natural communications between middle ear and inner ear
- **Bony Erosion**
  - Caused by cholesteatoma or atticoantral type of CSOM
  - Proteolytic enzymes produced lead to bony erosion, facilitating infection spread.
- **Venous Spread**
  - The sigmoid sinus is located posteriorly in the mastoid
  - Venous spread → Sigmoid sinus thrombosis (Lateral sinus thrombosis)
  - There is no arterial route of spread for infection from middle ear

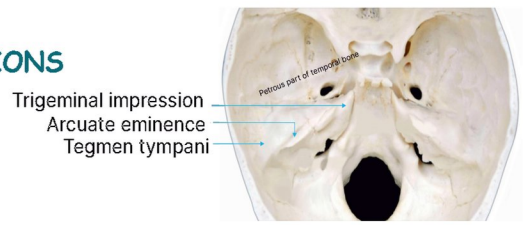
CLASSIFICATION OF COMPLICATIONS

00:01:30

Type	Complications
EXTRACRANIAL COMPLICATIONS	Mastoiditis; Labyrinthitis; Petrositis; Osteomyelitis; Septicaemia; Facial nerve palsy
INTRACRANIAL COMPLICATIONS	Meningitis; Extradural abscess; Subdural abscess; Brain abscess; Otitic hydrocephalus; Lateral sinus thrombosis

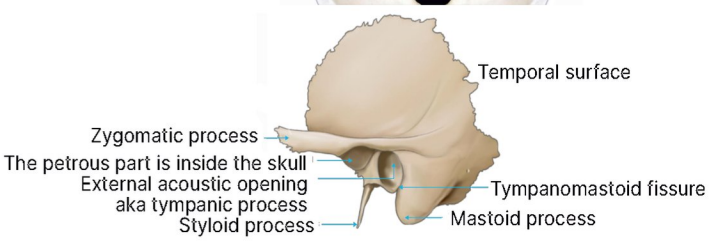
Feature	Acute Otitis Media	Chronic Otitis Media
Rule	The answer usually ends with **"-itis"**	The answer usually **does not end with "-itis"**
Most common extracranial complication	Mastoiditis	Post-auricular abscess
Most common intracranial complication	Meningitis	Intracranial abscess (Brain abscess)

ANATOMICAL BASIS OF EXTRACRANIAL COMPLICATIONS



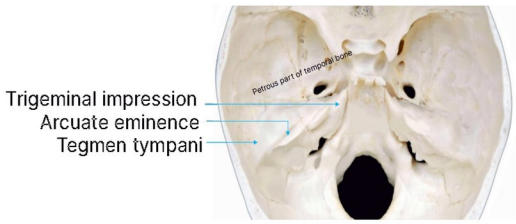
PARTS OF TEMPORAL BONE:

- Petrous part (leaf-like bone)
  - Contains inner ear
  - The internal auditory canal is present here
- Tympanic part
  - Forms part of external auditory canal
- Squamous part



- Tegmen tympani = roof of middle ear
- Mastoid part
  - Contains mastoid air cells

EXTRACRANIAL COMPLICATIONS:



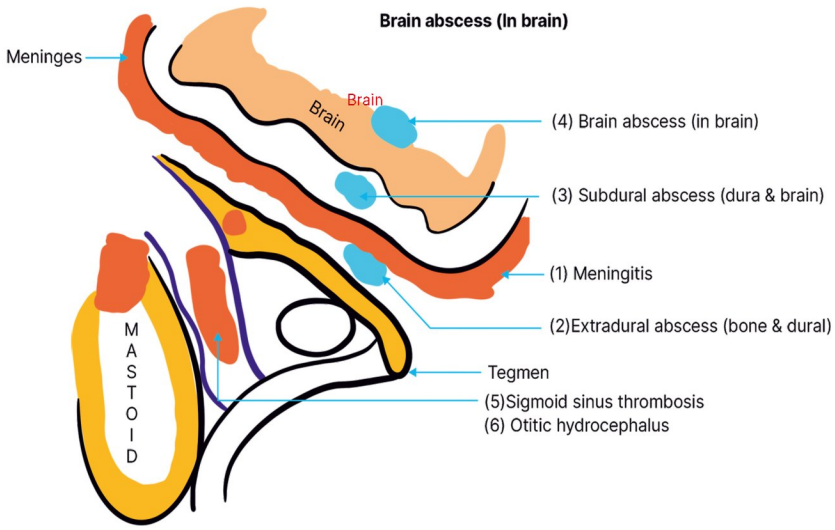
- Spread to Mastoid air cells → Mastoiditis
- Spread medially to Inner ear → Labyrinthitis (pus in inner ear)
- Spread medially to Petrous bone apex → Petrositis
- Spread to Squamous temporal bone → Osteomyelitis
- Blood spread → Septicaemia
- Spread to the facial nerve (travels from IAC → Inner ear → Middle ear → Mastoid) → Facial nerve palsy

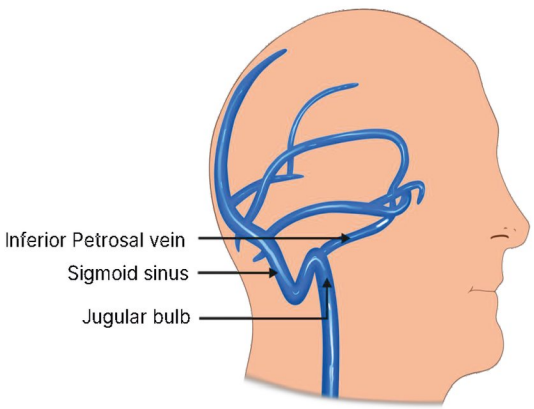
ROUTES OF INTRACRANIAL COMPLICATIONS

- Route 1 (Most Common):
  - Middle ear disease → Breakthrough Tegmen tympani (roof of middle ear) → Enters Middle cranial fossa → Contains Temporal lobe of brain → Most common site for intracranial complications
- Route 2 (Less Common):
  - Middle ear → Inner ear → Petrous bone → Cerebellum

INTRACRANIAL COMPLICATIONS

- Disease erodes Tegmen → Inflammation of meninges → Meningitis
- Pus between bone and dura → Extradural abscess
- Pus breaks through dura → lies between dura and brain parenchyma → Subdural abscess
- Pus goes inside brain → Brain abscess (Intracerebral abscess)
- Disease reaches Sigmoid sinus → Thrombosis of sigmoid sinus → Lateral sinus thrombosis
- Sigmoid sinus thrombosis → Cranial circulation cannot drain → Fluid retention in ventricles → Otitic hydrocephalus





EXTRACRANIAL COMPLICATIONS

00:07:20

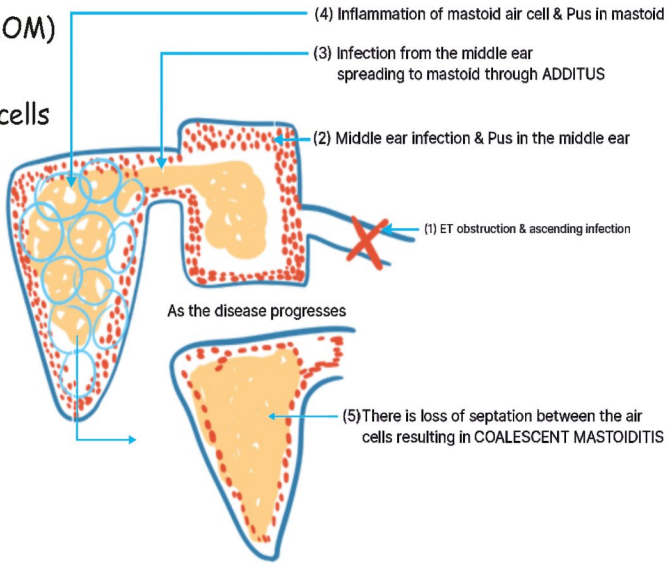
ACUTE MASTOIDITIS

- Most common extracranial complication of Acute Otitis Media
- Causative Organism: Streptococcus pneumoniae (same as AOM)

YOURWISH

PATHOPHYSIOLOGY:

- Eustachian tube blockage → Pus collects in middle ear (AOM) → Pus travels via Aditus → Reaches Mastoid antrum → Inflammation of mastoid air cells → Pus fills mastoid air cells → Septa between air cells break → Air cells merge into single pus-filled cavity → Coalescent mastoiditis

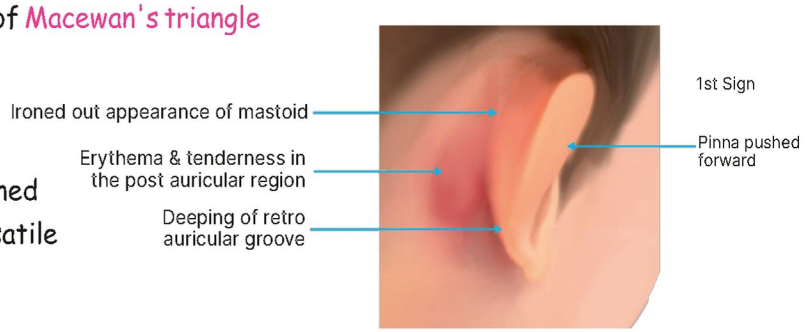


SYMPTOMS (EXAGGERATED ASOM SYMPTOMS):

- Pain (more severe, also behind pinna)
- Fever
- Hearing loss
- Ear discharge
  - profuse and persistent can be pulsatile
  - Lighthouse sign
- Tenderness positive

SIGNS:

- "Ironed out appearance" of mastoid
  - Skin is stretched, red, oedematous.
  - This is the first sign of acute mastoiditis
- Pinna Changes:
  - Pinna pushed outwards and forward
- Sagging of posterior meatal wall
- Tenderness on the mastoid region on the area of Macewan's triangle
- Retroauricular Groove:
  - Deepening of retro-auricular groove
  - Due to swollen, edematous mastoid
  - Gives apparent feeling that groove is deepened
- Discharge on otoscopy reflects light and is pulsatile
  - Pulsatile otorrhoea
  - Called Lighthouse sign
  - Also seen in ASOM (Non-specific sign)
- Specific sign (only in mastoiditis):
  - Reservoir sign
    → Pus in the external auditory canal (mucopurulent)
    → After removing the pus, the canal will immediately fill with pus again.
  - If it takes time to reappear, it is likely AOM rather than mastoiditis.



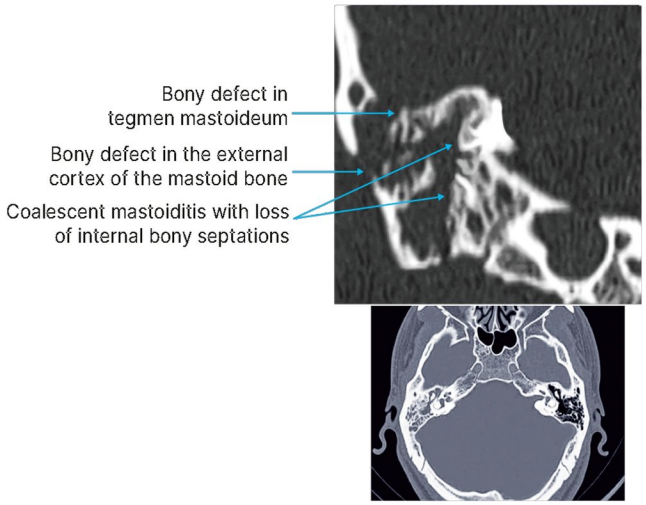
DIAGNOSIS:

- Clinical:
  - History and examination findings
    → Ironed out appearance
    → Pinna pushed outwards and forwards
    → Deepening of retro-auricular groove

YOURWISH

→ Reservoir sign positive

- Imaging:
  - X-ray or CT scan
- CT Findings:
  - Coalescence of air cells visible
  - Destruction/defect in roof of mastoid
  - Haziness in mastoid
  - Compare with opposite side: the diseased side is hazy (not black/air-filled)



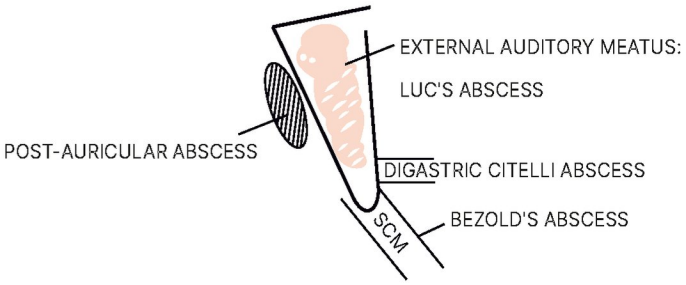
TREATMENT

- Surgery: Cortical Mastoidectomy
- Procedure:
  - Delineate the entire mastoid from:
    - → Roof (superiorly)
    - → Sigmoid sinus (posteriorly)
    - → Tip cells (inferiorly)
    - → Posterior wall of external auditory canal (anteriorly)
  - Remove all diseased air cells
- Medical: IV antibiotics



LOCAL ABSCESSES RELATED TO MASTOID

- **BEZOLD'S ABSCESS**
  - Pus breaks through the tip of the mastoid.
  - Goes into the sternocleidomastoid muscle
  - Present in upper part of neck
- **POST-AURICULAR ABSCESS**
  - MC abscess
  - Pus breaks through the outer cortex of the mastoid.
  - Presents as abscess behind the pinna
- **LUC'S ABSCESS**
  - Pus breaks into the external auditory meatus.
  - Memory aid: L follows M
    - → Luc's = Meatus
- **CITELLI'S ABSCESS (DIGASTRIC ABSCESS)**
  - At tip of mastoid = Posterior belly of digastric muscle
  - Abscess in digastric region
  - Memory aid: C follows D
    - → Citelli's = Digastric



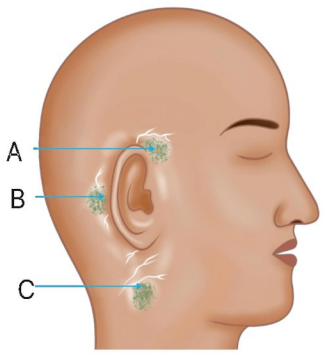


Image based

- Near zygomatic bone → Zygomatic abscess
- Behind pinna → Post-auricular abscess (MC)

- Upper part of neck in SCM → Bezold's abscess
- Near angle of mandible → Digastric (Citelli's) abscess
- In external auditory meatus → Luc's abscess

PETROSITIS

00:14:03

- Definition: Inflammation of petrous apex cells
- Anatomy:
  - Petrous bone = Leaf-like bone
  - At apex of petrous bone: Two cranial nerves
    → 5th cranial nerve (Trigeminal)
    → 6th cranial nerve (Abducens)

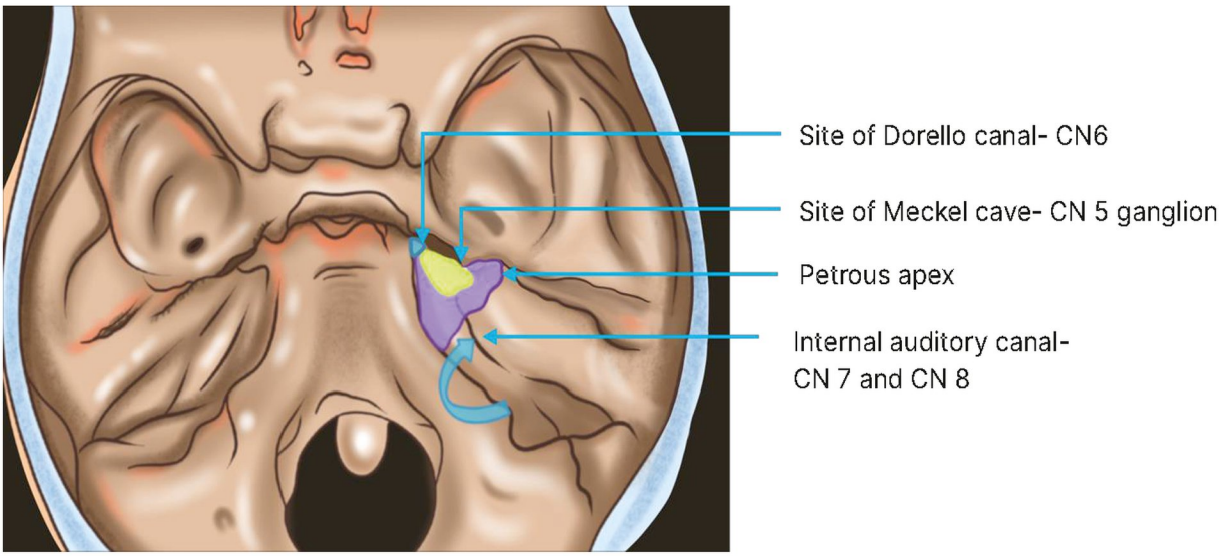
PATHOPHYSIOLOGY

- Infection spreads from middle ear to the petrous apex
- Pus/inflammation at petrous apex
- Involves 5th and 6th cranial nerves

CLINICAL FEATURES

GRADENIGO'S TRIAD

- Persistent Ear Discharge
  - Constant otorrhoea from middle ear disease
- Deep-Seated Retro-Orbital Pain
  - Due to 5th nerve involvement (Trigeminal nerve)
  - Pain around the eye
- Lateral Rectus Palsy (Diplopia)
  - Due to 6th nerve involvement (Abducens nerve)



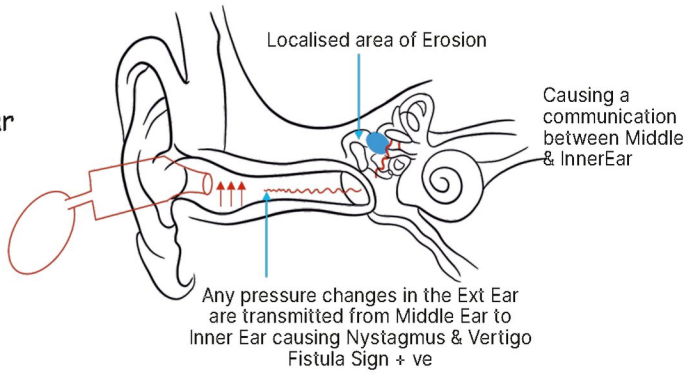
LABYRINTHITIS

00:15:18

- Definition: inflammation of the inner ear

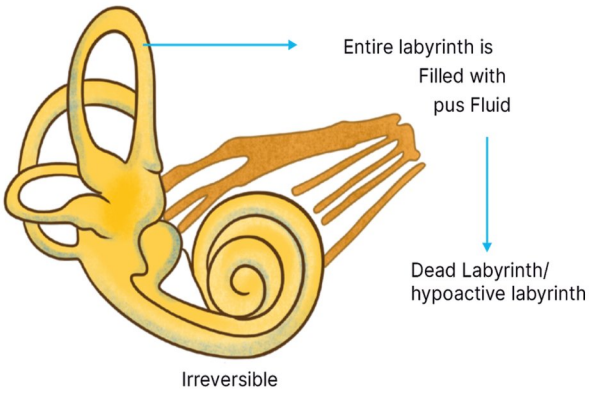
THREE FORMS:

CIRCUMSCRIBED LABYRINTHITIS



- Pathology:
  - Localized area of labyrinth involved
  - Small area of erosion between middle ear and inner ear
  - Communication formed between middle and inner ear
- Clinical Features:
  - Fistula sign: positive
  - Mechanism: Increased pressure in external auditory canal → Pressure reaches middle ear → Through the erosion area →infection reaches inner ear
  - Results in transient vertigo when pressure applied
- Prognosis: Localized, can be managed

SEROUS (DIFFUSE SEROUS) LABYRINTHITIS



- Pathology:
  - The entire labyrinth filled with Sterile fluid
  - No pus present
  - Liberation of bacterial toxins causes sterile inflammation
- Hyperactive labyrinth (irritated)
- Clinical Features:
  - Vertigo:
    → Single, long episode
    → Lasts few days to few weeks
  - Nystagmus:
    → towards the same side (diseased side)
    → Rule: Hyperactive labyrinth → Nystagmus towards same side
  - Sensorineural Hearing Loss (SNHL)
  - Tinnitus
  - NO fever (no pus = no toxic features)
- Prognosis:
  - Reversible condition
- With appropriate antibiotics
- Clear infection/disease
- Can reverse the condition

SUPPURATIVE (DIFFUSE SUPPURATIVE) LABYRINTHITIS

- Pathology:
  - The entire labyrinth filled with PUS
  - Purulent fluid throughout inner ear
- Hypoactive labyrinth (destructive)
- Clinical Features:
  - Vertigo:

YOURWISH

→ Severe vertigo
- Nystagmus:
  → Towards opposite side (away from diseased side)
  → Rule: Hypoactive labyrinth → Nystagmus towards opposite side
- Sensorineural Hearing Loss (SNHL)
- Tinnitus
- HIgh grade fever (pus = toxic features)
- Toxic symptoms present

- Prognosis:
  - Irreversible condition
  - Requires immediate attention
- Treatment:
  - Labyrinthine sedatives
  - Mastoid exploration (MRM)
  - Clear disease source
  - IV antibiotics
  - If not treated → Progresses to intracranial complications

Feature	Circumscribed Labyrinthitis	Diffuse Serous Labyrinthitis	Diffuse Suppurative Labyrinthitis
Pathology	Thinning or erosion of bony capsule of labyrinth	Diffuse inflammation of labyrinth without pus	Pus infection of labyrinth
Labyrinth status	Labyrinth exposed to external stimuli	Labyrinth inflamed and irritable	Loss of vestibular and cochlear functions
Reversibility	Reversible	Reversible	Irreversible
Fistula test	Positive	Negative	Negative
Nystagmus	Positive pressure → nystagmus to same side; Negative pressure → nystagmus to opposite side	Nystagmus to same side	Nystagmus to opposite side
Symptoms	Transient vertigo on applying pressure to tragus	Vestibular and cochlear symptoms	Toxic severe symptoms of serous labyrinthitis

KEY RULE FOR NYSTAGMUS:
- Hyperactive/Irritated labyrinth (Serous) → Nystagmus Towards same side
- Hypoactive/Destroyed labyrinth (Suppurative) → Nystagmus Towards opposite side

YOURWISH

YOURWISH

APPROACH TO PATIENT WITH AOM + COMPLICATIONS

Intratemporal complications of otitis media

Patient of Otitis Media presents with

Child with Fever, ear discharge, Postauricular edema and tenderness

Next step

On Examination



- Pinna pushed outward and downward
- Postauricular - Ironed out
- EAC -Reservoir sign, Sagging of posterosuperior meatal wall.
- Abscess following mastoiditis Postauricular (MC).
- Bezold's -In relation to SCM.
- Citelli's -In relation to posterior belly of digastric/ occipital region.
- Luc's -In relation to posterior wall of EAC.

Persistent Ear discharge + retroorbital pain (5th inv.) + Diplopia (6th palsy) - Gradenigo triad (due to abscess reaching Petrous Apex)

Diagnosis: Petrositis

Management: IV antibiotics + MRM + drainage of petrous air cell.

Mixed HL, Vertigo

Diagnosis:

- Labyrinthitis (erosion is seen MC of → lateral semicircular canal).

Management: Labyrinthine sedative IV antibiotics + if has followed chronic otitis media then MRM.

Deviation of face to the normal side



Diagnosis: Facial palsy

Management: MRM with facial nerve decompression.

Diagnosis: Mastoiditis (MC caused by-Pneumococcus)

Management: IV antibiotics + cortical/simple mastoidectomy Schwartze operation

INTRACRANIAL COMPLICATIONS

MENINGITIS

00:21:05

- Most common intracranial complication of AOM
- Causative Organism: Streptococcus pneumoniae (same as AOM)
- Clinical Features:
  - High grade fever
  - Projectile vomiting
  - Projectile headache
  - Altered sensorium/unconsciousness
  - Irritability
- Specific Signs:
  - **BRUDZINSKI'S SIGN** (Neck Sign):
    → Flexion of neck → Automatic flexion of hip and knee
    → Tests meningeal irritation
  - **KERNIG'S SIGN:**
    → Thigh flexed towards abdomen
    → Attempt to extend leg → Pain
    → Tests meningeal irritation
    → Both signs = Specific for meningitis
- Diagnosis:
  - Gold Standard: Lumbar puncture

- Send CSF for analysis
- Based on CSF analysis → Treatment decided

- Treatment:
  - Based on CSF findings
  - IV antibiotics with mastoid exploration

EXTRADURAL ABSCESS

- Location:
  - Between dura and bone
  - Outside the dura (that's why "extra-dural")
- Characteristic Feature:
  - Usually Silent
  - May not present with dramatic symptoms
- Treatment:
  - Modified Radical Mastoidectomy
  - abscess is drained with IV antibiotics

SUBDURAL ABSCESS

- Location:
  - Between dura and brain parenchyma
- Characteristic Feature:
  - Very Rare complication
  - Not a common complication
- Clinical Features:
  - Fever
  - Headache
  - Projectile vomiting (raised ICP)
  - Some localizing signs
- Treatment:
  - IV antibiotics and drugs which reduce ICP
  - Mastoid exploration

BRAIN ABSCESS (INTRACEREBRAL ABSCESS)

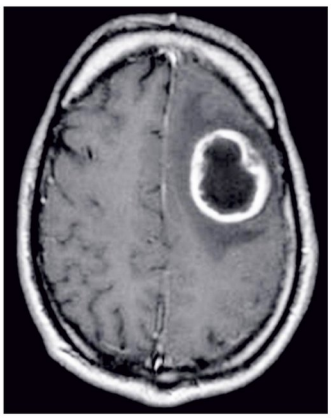
NEET PG 2021

- Most Common Type: Intracerebral abscess (Brain abscess)
  - NOT extradural
  - NOT subdural
- Location:
  - Temporal lobe is most commonly involved
  - Disease spreads from middle ear → Tegmen → Temporal lobe
- Two Types of Features:
  - Due to raised intracranial tension
    → Headache,Projectile vomiting,Papilloedema,Drowsiness,Confusion,Stupor Bradycardia
  - Localising features
    → Temporal lobe abscess
      - Speech cortex → Aphasia

- Motor cortex → Hemiplegia
- Sensory cortex → Sensory loss
- Visual cortex → Hemianopia

→ If Cerebellar Abscess:

- Ataxia
- Dysmetria
- Dysdiadochokinesia



- Diagnosis: MRI with Contrast
  - Characteristic Finding: Ring sign
    → Central dark area (necrotic pus)
    → Surrounded by enhancing ring (capsule)
    → Ring sign = Brain abscess
- Treatment:
  - Medical Management:
    → IV antibiotics, dexamethasone
    → Medications to reduce intracranial pressure: Mannitol
  - Neurosurgical Drainage:
    → Drain the abscess first
  - Ear Surgery:
    → Modified Radical Mastoidectomy for CSOM/Cholesteatoma
- MC type of brain abscess: Intracerebral abscess
- Lobe involved is temporal
- Ring sign on MRI see in intracerebral abscess

LATERAL SINUS THROMBOSIS (SIGMOID SINUS THROMBOSIS)

00:24:10

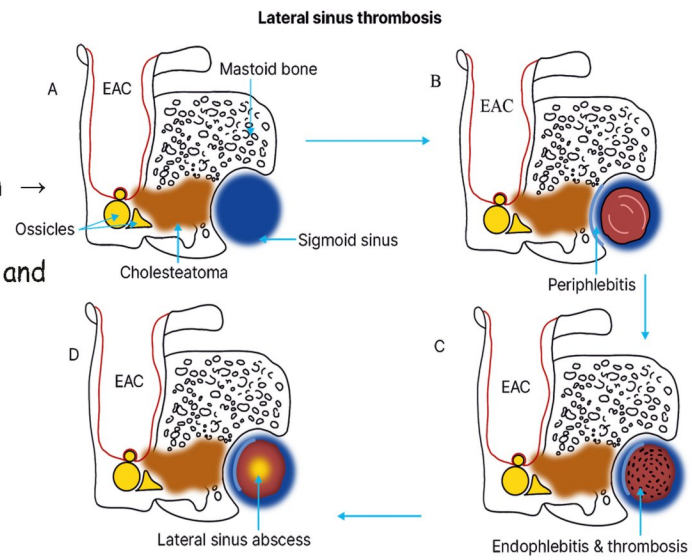
FMGE 2018, 2020

ANATOMY

- The sigmoid sinus forms posterior border of mastoid
- Part of venous drainage of brain

PATHOPHYSIOLOGY

- Disease spreads from Middle ear → Mastoid air cells
- Involves outer wall of sigmoid sinus → Periphlebitis
- Then involves inner wall → Endophlebitis
- Periphlebitis/Endophlebitis → Platelet aggregation → Thrombus formation (infective thrombus) →Later forms abscess and blocks sigmoid sinus



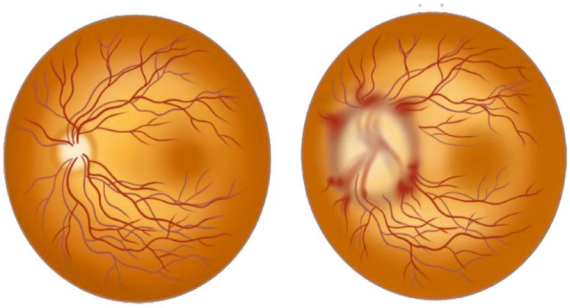
CLINICAL FEATURES:

- HECTIC PICKET FENCE FEVER:
  - Constant fever
  - Does NOT touch baseline
  - Characteristic pattern
  - Due to emboli release to circulation

YOURWISH

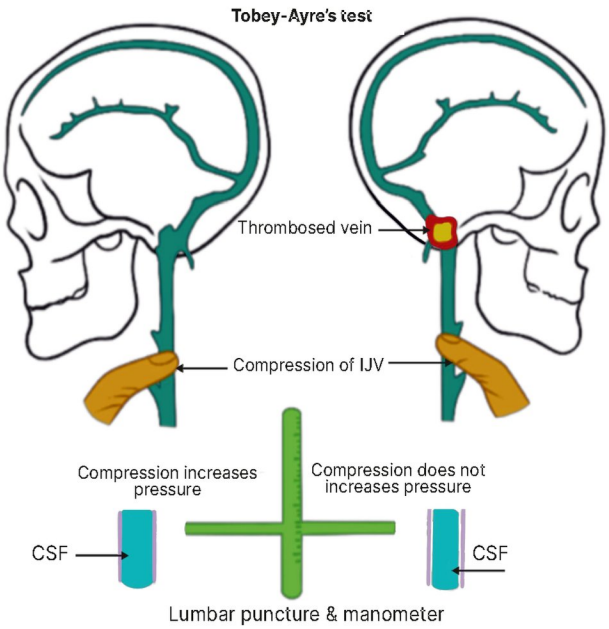
- **GRIESINGER SIGN:**
  - Edema and bluish discoloration of mastoid
  - Due to improper drainage of mastoid emissary veins
- **RAISED INTRACRANIAL PRESSURE:**
  - Presenting with papilloedema
- **TOBEY-AYER TEST (Historical):**
  - Not done now
  - Was used in past
  - Compression of IJV on the healthy side produces rise in CSF pressure
- **CROW BECK TEST(Historical):**
  - Not done now
  - Was used in past
  - Compression of IJV on the healthy side produces engorgement of retinal veins





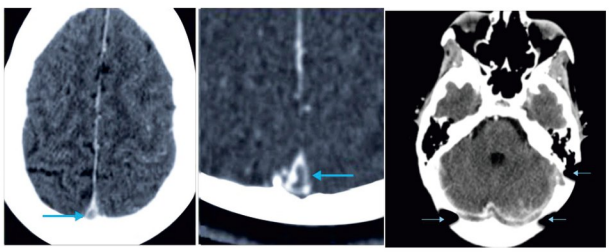
Normal retina

Retina with papilledema



DIAGNOSIS:

- Modern Investigation: Contrast-Enhanced CT/MRI
- Characteristic Finding: delta sign or empty delta sign
  - Diagnostic of sigmoid sinus thrombosis



COMPLICATIONS:

OTITIC HYDROCEPHALUS:

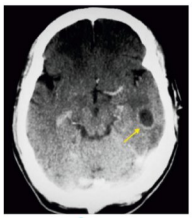
- Sigmoid sinus thrombosed
- Cranial circulation cannot drain
- Obstruction to venous drainage
- Fluid retention in ventricles
- Dilatation of ventricles = Hydrocephalus
- C/F
  - Headache,Papilloedema,Projectile vomiting
- Diagnosis by CT: dilated ventricles

YOURWISH

- Rx
  - Medical treatment to reduce ICP
  - VP Shunt (Ventriculoperitoneal shunt) to redirect CSF

Patient of Otitis Media presents with

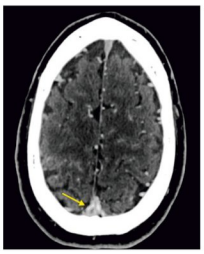
- Nominal aphasia, Homonymous supra-quadrantanopia, Headache, Seizures, Contralateral hemiparesis
- CT/MRI-Ring sign



Diagnosis : Temporal lobe abscess

Management: Abscess drainage + IV antibiotics + MRM when the patient is stable

- Picket fence fever , tenderness and edema over the mastoid k/a Griesinger's sign (mastoid emissary vein thrombosis).
- Tenderness on the neck over the jugular vein area
- Tobey Ayer test/(Queckenstedt's test) -Positive (increased CSF pressure on compression of normal jugular vein)
- Crowe Beck sign (retinal vein engorgement/papilledema) on IJV compression of normal side-present.
- **CT/MRI:** Delta sign or Empty triangle sign



Diagnosis : Lateral sinus thrombophlebitis

Management : IV antibiotics + MRM

Fever headache, vomiting, drowsiness, neck rigidity and irritability, Kernig's, Brudzinski's sign (+)

Diagnosis: MeningitisMC intracranial complication (through cochlear aqueduct).

Management: IV antibiotics + MRM when patient stable.

COMMON ERRORS TO AVOID

- Routes of Spread
  - Three routes:
    - → Natural communications (oval window, round window)
    - → Bone destruction (proteolytic enzymes)
    - → Venous spread (sigmoid sinus)
  - No arterial route of spread
- Pulsatile Otorrhea (Lighthouse Sign)
  - It is a Non-specific sign
  - Lighthouse sign seen in Both:
    - → AOM (Acute Suppurative Otitis Media)
    - → Acute Mastoiditis
  - Specific Sign for Mastoiditis:
    - → Reservoir sign Pus removed → Immediately refills
- Gradenigo's Triad
  - Seen in Petrositis
  - At petrous apex: 5th and 6th cranial nerves
  - Triad:
    - → Persistent ear discharge
    - → 5th nerve → Retro-orbital pain
    - → 6th nerve → Lateral rectus palsy (diplopia)

- Lateral Sinus Thrombosis Signs
  - Griesinger Sign:
    → Edema and bluish discoloration of mastoid
    → Due to improper venous drainage
  - Tobey-Ayer Test: (Historical)
  - Queckenstedt Test: (Historical)
  - Delta Sign/Empty Delta Sign:On contrast CT/MRI

MCQ

Q. A 4 year old boy was suffering ear infection and tonic spasms of limbs, with fever and neck rigidity. Which of the following is the likely diagnosis? INICET May 2025

A. Meningitis
B. Bezold
C. Sigmoid sinus thrombophlebitis
D. Tetanus

Answer: D

Q. A patient with CSOM was presented to the OPD with seizures. On examination and evaluation, homonymous hemianopia and epilepsy were present. Identify the condition with the CT scan given below?

A. Temporal abscess
B. Cerebellar abscess
C. Extradural abscess
D. Meningitis

Answer: A

Q. Which of the following are symptoms of Gradenigo syndrome?

A. Ear discharge, retro-orbital pain, and diplopia
B. Nasal discharge, retro-orbital pain, and diplopia
C. Ear discharge, retro-orbital pain, and headache
D. Retro-orbital pain, Diplopia, and hearing loss

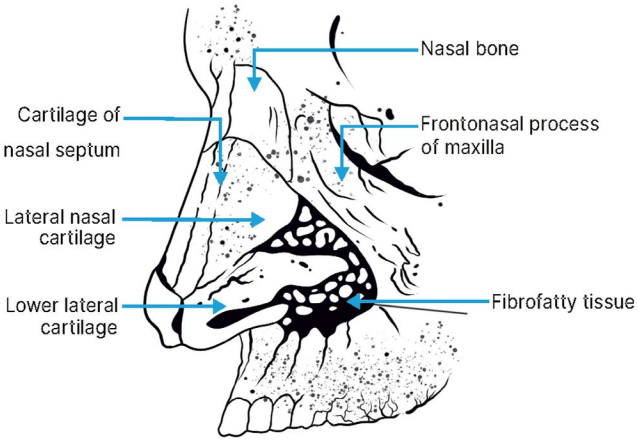
Answer: A

12. RHINOLOGY PART-1

ANATOMY OF THE NOSE

00:02:11

External Nose Anatomy



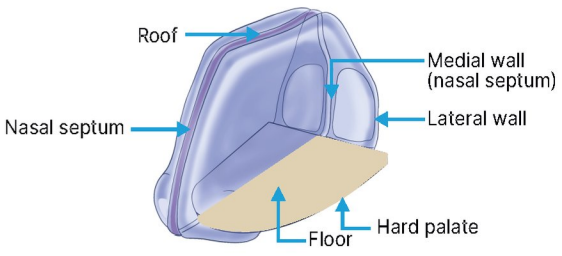
Bony pyramid	Cartilaginous pyramid
• upper 1/3rd **Paired bones** • Nasal bone • Ascending process of the maxilla **Unpaired bones** • Frontal bone	• lower 2/3rd **(PYQ)** **Paired cartilage** • Upper lateral cartilage • Lower lateral cartilage • Small sesamoid cartilage **Unpaired cartilage** • Septal cartilage (comes from below and gives support to the dorsum of the nose)

Nasion	• Part of the bone where the frontal bone meets the nasal bone in the midline
Rhinion	• The nasal bone unites with the upper lateral cartilage in the midline (PYQ)
Limen Nasii	• Lower border of the upper lateral cartilage; this is the site of incision in internal rhinoplasty • It forms the boundary of the internal nasal valve (the narrowest portion of the nasal cavity)

Important Information

- Site of incision for external rhinoplasty- incision on the outside of the columella of the nose)

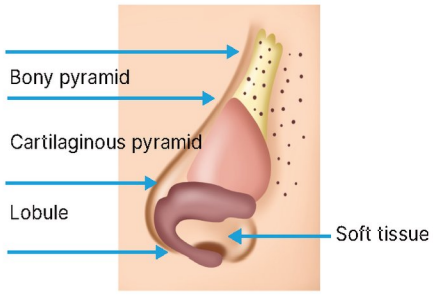
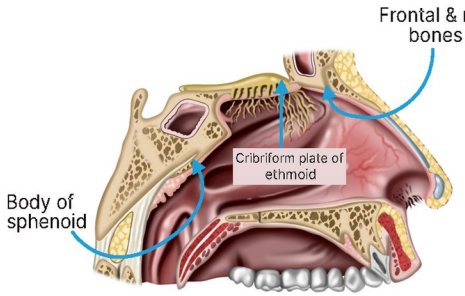
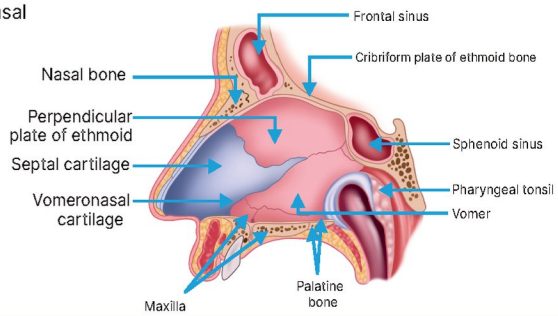
Internal Nose Anatomy



Nasal Cavity

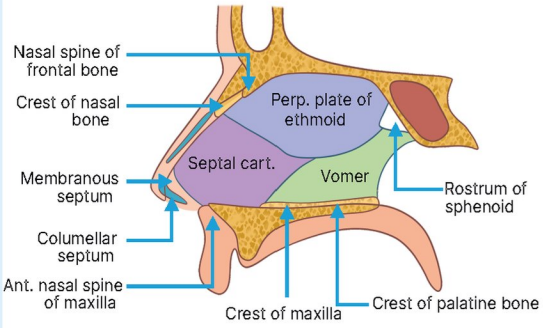
- The nasal cavity proper has got 6 walls

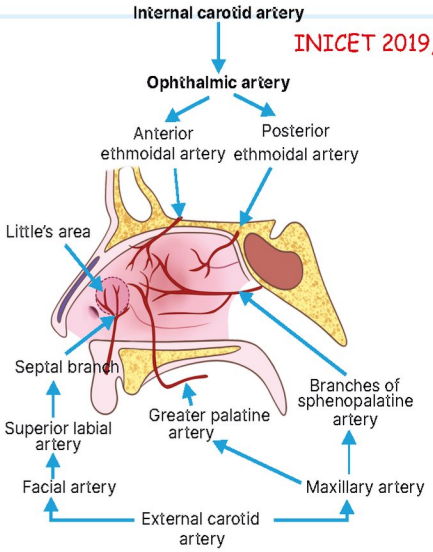
YOURWISH

Inlet- Vestibule	• The only area in the nose lined by stratified squamous epithelium with hair follicles and sebaceous glands (the rest is pseudostratified ciliated columnar epithelium) • Folliculitis is most common in this region • Contains an internal nasal valve • **Formed by** ○ Medially- Septum ○ Superiorly- Lower border of the upper lateral cartilage ○ Laterally- Inferior turbinates   
Outlet- Choana	• Communication between the nose and the nasopharynx
Superiorly- Roof	• Has specialised sensory epithelium- the olfactory epithelium (perceives smell) • Separates nasal cavity from the cranial cavity • Formed by ○ Anterior 1/3rd- nasal bone and frontal bone ○ Middle 1/3rd- cribriform plate of ethmoid bone ○ Posterior 1/3rd- sphenoid bone
Inferior- Floor	• Separates the nasal cavity from the oral cavity • Formed by ○ Crest of the maxilla ○ Crest of the palatine bone
Medially- Septum	• Separates the right and left halves of the nose • Made of cartilaginous and bony parts • Cartilage- septal cartilage/quadrangular cartilage/quadrilateral cartilage • Bone- perpendicular plate of the ethmoid and vomer ○ Other small contributions- Nasal spine of the frontal bone, Rostrum of the sphenoid bone, & Crest of the maxilla and the palatine bone **Blood Supply to the septum** • Internal carotid artery- Upper half • Anterior ethmoidal artery- Anterosuperior portion of the septum • Posterior ethmoidal artery- Posterosuperior portion of the septum • External carotid artery- Lower half • Superior labial artery- Anteroinferior portion of the septum • Sphenopalatine and greater palatine artery- Posteroinferior portion of the septum

YOURWISH

YOURWISH



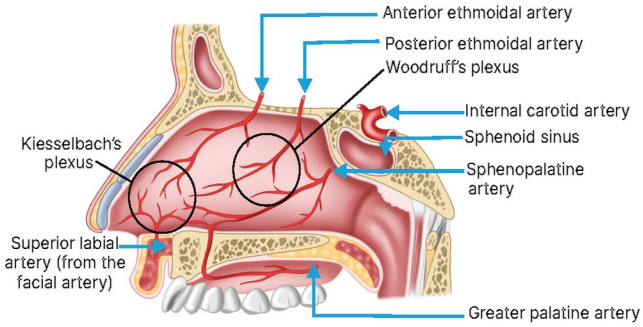


INICET 2019, FMGE 2021

Laterally- Lateral wall of the nose	• Has bony projections called turbinates

Kiesselbachs And Woodruffs Plexus

00:13:40



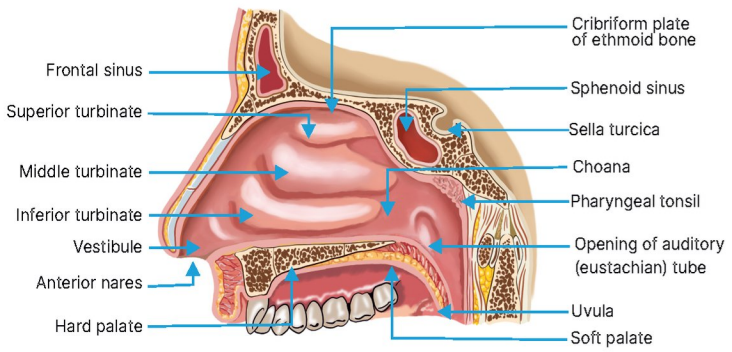
- Kiesselbach's (Little's Area)- An arterial plexus located in the anteroinferior portion of the septum, where anastomosis of all the arteries of the septum except the posterior ethmoidal artery
- Woodruff's venous plexus is at the posterior end of the inferior turbinate. Venous anastomosis causes posterior epistaxis

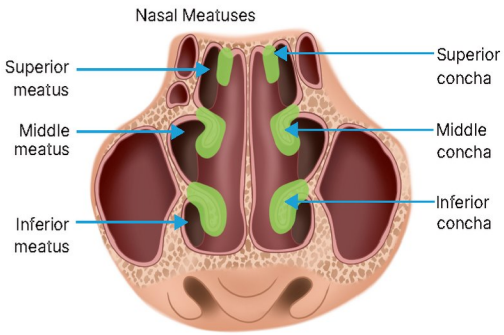
Lateral Wall Of Nose

00:15:10

INICET 2020, 2021

- 3 bony projections (turbinates/chonca)
- Meatus (spaces between bony projections)





Inferior meatus	• Receives the opening of the nasolacrimal duct guarded by the valve of Hasner
Middle meatus	• Receives the opening of the anterior group of sinuses (frontal, maxillary, and anterior ethmoidal sinuses) • Area to operate for FESS, sinusitis, and polyps **Uncinate process** • A boomerang-shaped bone is present anteriorly. • Behind it is the bulla ethmoidalis. • The space between the uncinate and bulla is the hiatus semilunaris (receives drainage of the frontal and anterior ethmoidal sinus), which leads inferiorly to the infundibulum (receives the drainage of the maxillary sinus) **Osteomeatal complex / Picardy Circle** • The final common point of drainage of all 3 sinuses is from the frontal, anterior ethmoidal, and maxillary sinuses.
Superior meatus	• Receives the opening of the posterior ethmoid

Note: The sphenoid sinus drains into the sphenoethmoidal recess

MCQs

00:19:43

Q. The external nose is formed from

a. 3 paired and 3 unpaired cartilages
b. 3 paired and 1 unpaired cartilage
c. 3 paired and 4 unpaired cartilages
d. 1 paired and 1 unpaired cartilage

Ans: **c**

Q. Osseocartilaginous junction on the dorsum of the nose is

a. Nasion
b. Rhinion
c. Columella
d. Glabella

Ans: b.

Q. The ethmoid bone forms all except

a. Superior turbinate
b. Inferior turbinate
c. Middle turbinate
d. Uncinate process

Ans: b.

Q. All the structures are present on the lateral wall of the nose except

a. Bulla
b. Vomer
c. Hassner's valve
d. Aggar nasi

Ans: b.

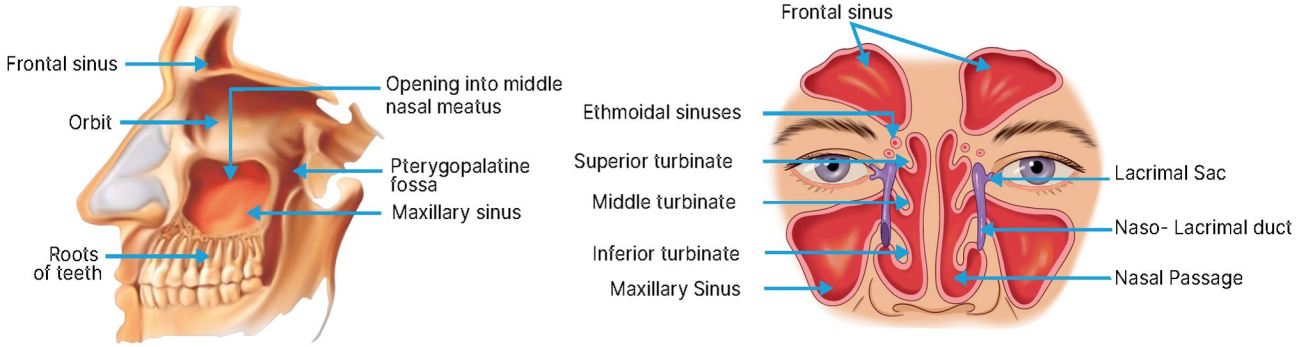
PARANASAL SINUSES

00:22:18

FMGE 2023

- 4 sinus groups

Maxillary Sinus

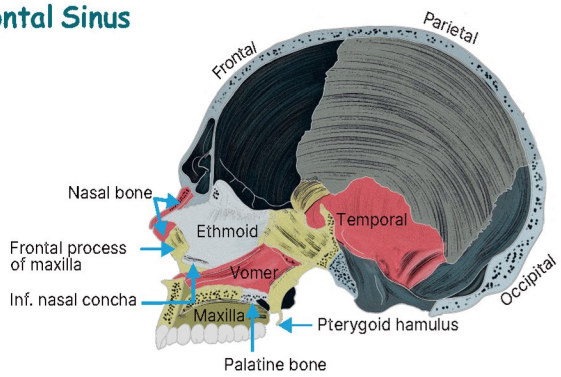


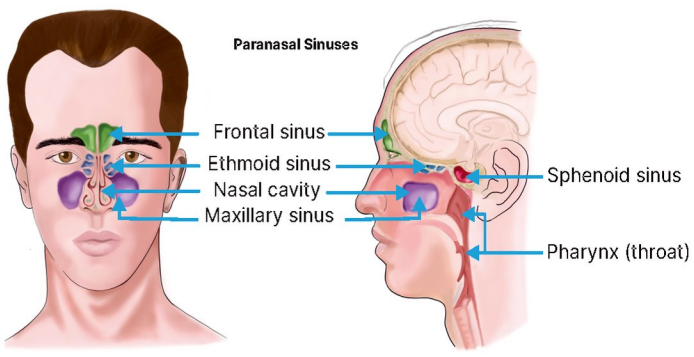
- AKA **Antrum of Highmore**
- The first sinus to develop and be visible radiologically
- Largest sinus
- MCC of sinusitis in adults

Boundaries

- Base- Separates the maxillary sinus from the lateral wall of the nose
- Apex- At the level of the zygoma
- Roof- Formed by the floor of the orbit (# of floor of orbit results in a tear drop sign)
- Floor- Related to the second premolar and first molar tooth
- Anterior wall- Skin and subcutaneous tissue
- Posterior wall- Pterygopalatine fossa, which contains the internal maxillary artery that divides into the sphenopalatine artery (through the sphenopalatine foramen) and the greater palatine artery (through the greater palatine foramen)

Frontal Sinus



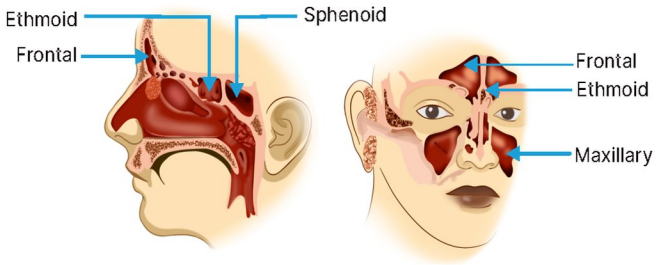


YOURWISH

- Last sinus to develop
- Absent at birth, therefore not visible radiologically at birth
- The frontal sinus is an air-filled cavity within the frontal bone.
- Pyramidal in shape, with anterior and posterior walls
  - The anterior wall- Formed by the anterior table of the frontal bone.
  - Posterior wall- Formed by the posterior table of the frontal bone.
- Chronic frontal sinusitis can erode the frontal sinus posteriorly and affect the frontal lobe of the brain, causing a frontal lobe abscess, causing personality changes, features of raised ICP, headaches, projectile vomiting, and an altered sensorium.
- The floor of the frontal sinus is related to the roof of the orbit, causing orbital complications
- Medially, the right and left frontal sinuses are separated by the interfrontal septum

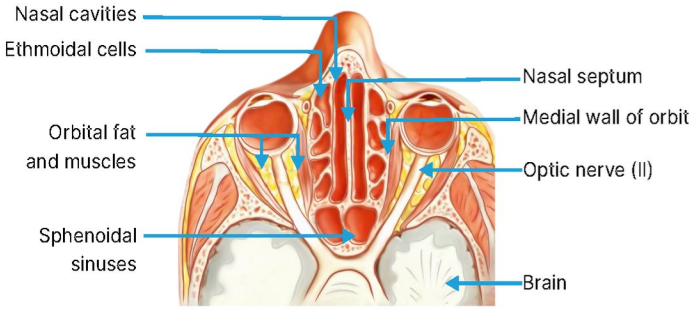
Ethmoid Sinus

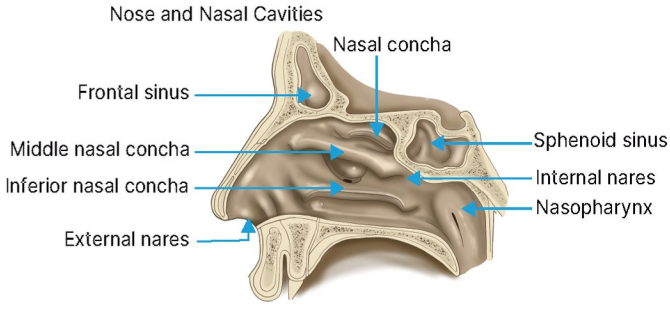
FMGE 2020, 2022, AIIMS 2020



- Most pneumatized sinuses at birth
- Sandwiched between the frontal sinus anteriorly and the sphenoid sinus posteriorly
- Roof- A thin plate separates from the anterior cranial fossa, called the fovea ethmoidalis.
- Medially related to the middle turbinate. The anterior and posterior ethmoids are divided by the basal lamella of the middle turbinate
- Laterally by the medial wall of the orbit
- Air cells
  - Agar nasii- Anteriormost ethmoidal cell
  - Bulla ethmoidalis- The biggest ethmoidal air cell
  - Haller cell- Ethmoidal cell in relation to the floor of the orbit.
  - Onodi cells- Posterior ethmoidal cells extending into the sphenoid sinus. May have the optic nerve and internal carotid artery

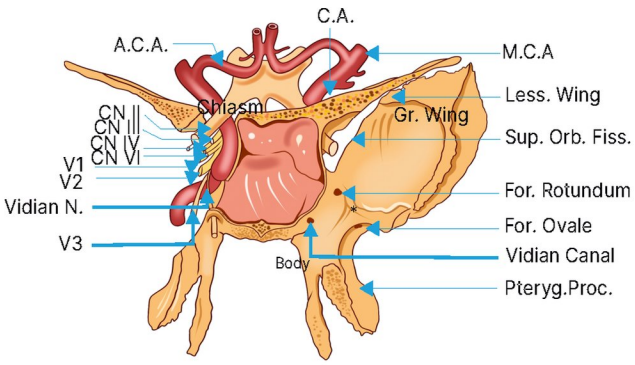
Sphenoid Sinus





YOURWISH

YOURWISH

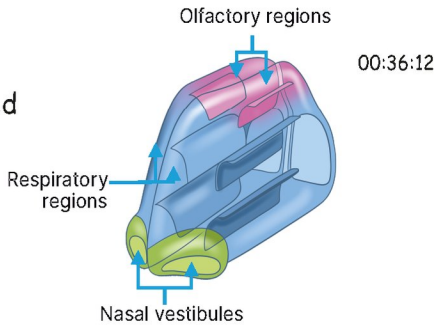


- The posterior-most sinus of the sphenoid bone
- Sella turcica- Depression in the roof of the sphenoid sinus that lodges the pituitary gland

Relations Of The Sphenoid Sinus

- Anterior- Posterior ethmoidal cells
- Posterior- Sphenoid bone.
- Lateral- Optic nerve, cavernous sinus (CN 3, 4, V1, V2, 6, ICA)
- Floor- Foramen rotundum above and vidian canal below.

LINING MEMBRANE OF NOSE



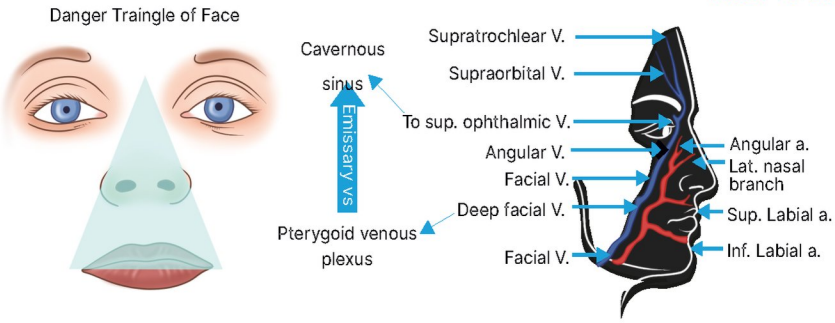
00:36:12

- Schneidunedian membrane/respiratory epithelium/Pseudostratified ciliated columnar epithelium lines the entire nasal cavity
- Exceptions
  - Vestibule- Stratified squamous columnar epithelium with hair follicles
  - Roof of the nose- Olfactory epithelium

DANGEROUS TRIANGLE OF NOSE

00:37:03

NEET PG 2018, 2020, FMGE 2021, 2023



- Area bounded between the nasion and the angle of the lips, as there is a very high possibility of infection going retrograde, which causes cavernous sinus thrombosis
- Pterygoid plexus of veins - as the veins are valveless, infections travel retrograde to the cavernous sinus and are therefore better treated with antibiotics

One Liners:

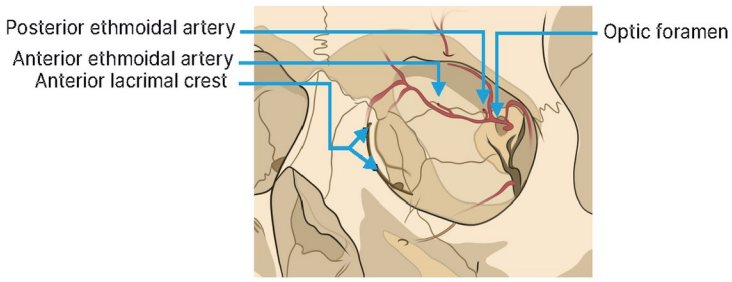
1. Turbinate, which is not a part of the ethmoid bone: Inferior turbinate
2. Sinus draining into the hiatus semilunaris: Frontal sinus, and the anterior ethmoidal sinus
3. Sinus draining into the infundibulum: Maxillary sinus
4. Sinus draining in the superior meatus: Posterior ethmoid
5. The sphenoid sinus drains into the sphenoethmoidal recess
6. Sinus absent at birth: Frontal sinus

7. The sinus whose development coincides with the development of secondary dentition is the maxillary sinus
8. The earliest sinus to develop and become visible radiologically is the maxillary sinus
9. Largest sinus: maxillary sinus
10. The last sinus to develop was the frontal sinus
11. The most common sinusitis in children is ethmoidal sinusitis
12. The most common sinusitis in adults maxillary sinus

EXTRAEDGE

00:40:30

Medial Wall Of Orbit



- 3 important structures on the medial wall:
  - Anterior ethmoidal foramen
  - Posterior ethmoidal foramen
  - Optic canal
- The anterior lacrimal crest is 24 mm from the anterior ethmoidal foramen
- 12 mm between the anterior ethmoidal and posterior ethmoidal foramina
- 6 mm posterior to the ethmoidal foramen to the optic canal



Nasociliary/anterior Ethmoidal nerve block 1 cm Above medial canthus



Infratrochlear nerve block -just medial to medial end of eyebrow



External nasal nerve block





- Nerve block given medially- Nasociliary block - nerve that gives rise to the anterior ethmoidal nerve
- Nerve block given to the dorsum of the nose- Anterior ethmoidal foramina block
- Greater palatine block- Given near the 3rd molar

MCQs

00:43:00

Q. The frontonasal duct opens into:

a. Inferior meatus
b. Middle meatus
c. Superior meatus
d. Inferior turbinate

Ans: b.

Q. Paranasal sinus openings in the middle meatus are

a. Maxillary
b. Anterior ethmoid
c. Posterior ethmoid
d. Frontal
e. Sphenoid

Ans: a, b & d

Q. The maxillary sinus opens into the middle meatus at the level of

a. Hiatus semilunaris
b. Bulla ethmoidalis
c. Infundibulum
d. None of the above

Ans: c

Q. Quadrilateral cartilage is attached to all except

a. Ethmoid
b. Vomer
c. Sphenoid
d. Maxilla

Ans: c

Q. The nasal valve is formed by

a. Upper lateral cartilage
b. Lower lateral cartilage
c. Lower end of the upper lateral cartilage
d. Upper end of the lower lateral cartilage

Ans: c

Q. All these structures are found in the lateral nasal wall except the following:

a. Superior turbinate
b. Vomer
c. Agger nasi
d. Hasner's vale

Ans: b

Q. The nasal valve is formed by all except:

a. Septum
b. Middle turbinate
c. Lower end of the upper lateral cartilage

YOURWISH

d. Inferior turbinate

Ans: b

Q. Onodi cells and Haller cells are seen in relation to
a. Optic nerve and floor of the orbit
b. Optic nerve and frontal sinus
c. Optic nerve and ethmoidal air cells
d. Orbit, chiasma, and nasolacrimal duct

Ans: a

Q. The osteomeatal complex (OMC) connects the following:
a. Nasal cavity with maxillary sinus
b. Nasal cavity with sphenoid sinus
c. The two nasal cavities
d. Ethmoidal sinus with ethmoidal bulla

Ans: a

Q. Which of the following is known as the fourth turbinate?
a. Superior turbinate
b. Aggernasi
c. Supreme turbinate
d. Bulous turbinate

Ans: a

Q. Ethmoid bone forms all except:
a. Superior turbinate
b. Middle turbinate
c. Inferior turbinate
d. Uncinate process

Ans: c

Q. During inspiration, the main current of airflow in a normal nasal cavity is through the
a. The middle part of the cavity in the middle meatus is in a parabolic curve
b. The lower part of the cavity in the inferior meatus is in a parabolic curve
c. Superior part of the cavity in the superior meatus
d. Through the olfactory area

Ans: a

YOURWISH

Q. A 19-year-old man is brought to the emergency department after a motor vehicle collision. On examination, the patient has a swollen face with closed eyes. An X-ray confirms a fracture involving the largest paranasal sinus. Which of the following sinuses is most likely involved?

a. Frontal
b. Ethmoid
c. Sphenoid
d. Maxillary

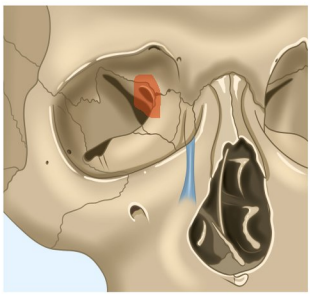
Ans: d

Q. The location of the woodruff plexus is

a. Posterior end of the middle turbinate
b. Anterior end of septum
c. Posterior end of the inferior turbinate
d. Posterior end of the superior turbinate

Ans: a

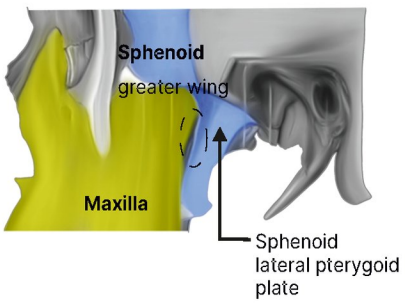
Q. The foramen shown is



a. Superior orbital fissure
b. Inferior orbital fissure
c. Optic canal
d. Anterior ethmoidal canal

Ans: c

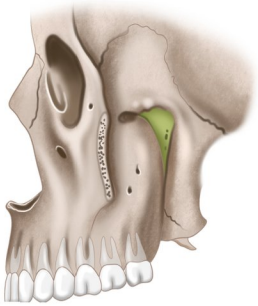
Q. The foramen shown is



a. Sphenopalatine foramen
b. Greater palatine foramen
c. Inferior orbital fissure
d. None

Ans: a

Q. The artery that is present in this fossa is



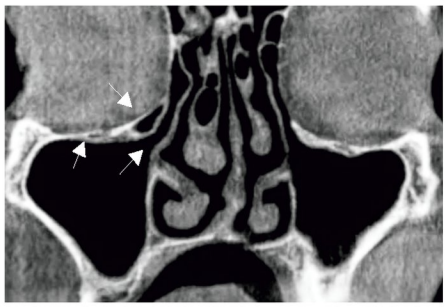
a. Ethmoidal artery
b. Internal maxillary artery
c. External carotid artery
d. Facial artery

Ans: b

Q. The arrow represents

a. Aggar nasi
b. Onodi cell

c. Haller cell
d. Billa ethmoidalis



Ans: c

Q. The direction of the nasolacrimal duct is
a. Downwards, backwards, and medially
b. Downwards, backwards, and laterally
c. Downwards, forwards, and medially
d. Downwards, forwards, and laterally

Ans: b

Q. Which type of sphenoid is a contradiction for the transnasal endoscopic approach to the pituitary gland?
a. Conchal
b. Presellar
c. Sellar
d. There is no contraindication

Ans: a

Q. The ciliary movement rate of the nasal mucosa is
a. 1-2 mm/min
b. 2-5 mm/min
c. 5-10 mm/min
d. 10-12 mm/min

Ans: c

Q. Odour receptors are present in
a. Olfactory epithelium
b. Olfactory tract
c. Amygdala
d. Olfactory bulbs

Ans: a

Q. The function of the mucociliary action of the upper respiratory tract is
a. Temperature regulation
b. Increased velocity of inspired air
c. Traps pathogenic organisms in inspired air
d. Has no physiological role

Ans: c

Q. Haller cells are present in

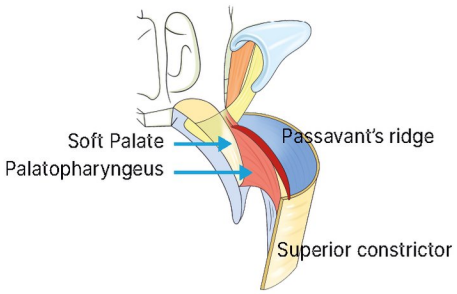
a. Roof of the maxillary sinus
b. Floor of the maxillary sinus
c. Posterior wall of the sphenoid sinus
d. Lateral wall of the sphenoid sinus

Ans: a

Q. A 7-year-old child with a recurrent cold and cough was diagnosed to have adenotonsillar hypertrophy. Parents complained of a nasal intonation of voice. This condition is,

a. Rhinolalia aperta
b. Rhinolalia clausa

Ans: b

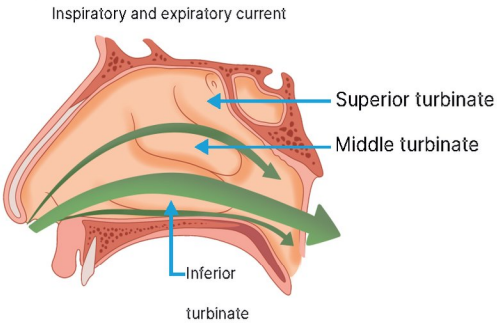


Passavant's Ridge

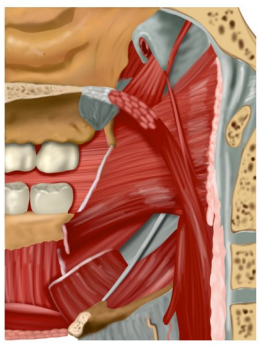
- Formed by the fibers of the palatopharyngeous muscle.
- Clinical Significance: If these are incompetent, they could increase vocal resonance by increasing airflow, leading to rhinolalia aperta.

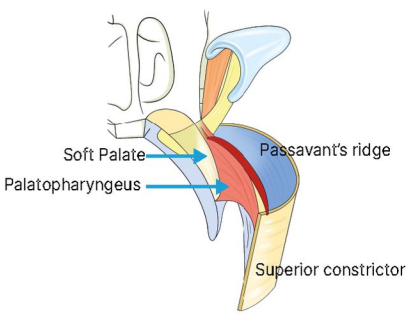
Inspiratory and expiratory current

- Maximum inspiratory current goes in a parabolic curve in the middle meatus



Passavant's Ridge





- Formed by the fibers of the palatopharyngeous muscle.
- Clinical Significance: If these are incompetent, they could increase vocal resonance by increasing airflow, leading to rhinolalia aperta.

YOURWISH

One Liners:

1. Hyposmia is a decreased perception of smell
2. Anosmia is the lack of perception of smell
3. Presbyosmia is an age-related decrease in the perception of smell
4. Paraosmia (Cacosmia or dysosmia) is an altered perception of smell
5. Phantosmia is the perception of smell even without a stimulus

Q. Oroantral fistula commonly occurs following the extraction of which tooth

a. Incisor
b. Canine
c. Premolar
d. 3rd Molar

Ans: c

Q. Pneumatization patterns of the sphenoid are all except the following:

a. Pre-seller
b. Post-seller
c. Concha bullosa
d. Conchal

Ans: c

Q. The roof of the nose is formed by all except:

a. Nasal process of the frontal
b. Cribriform plate of the ethmoid
c. Sphenoid
d. Palatine process of the maxilla

Ans: d

Q. The Eustachian tube is located at

a. Posterior to the superior turbinate
b. Posterior to the middle turbinate
c. Posterior to the inferior turbinate
d. At the sphenoethmoidal recess

Ans: c

Q. Which turbinate is seen only on nasal endoscopy?

a. Superior
b. Middle
c. Inferior
d. Eustachian tube

YOURWISH

Ans: a

Q. The Vidian nerve is

a. Auricular branch of vagus
b. Nerve of the pterygoid canal
c. Tympanic branch of the glossopharyngeal nerve
d. Lesser petrosal nerve

Ans: b

Q. The following are used to assess nasal mucociliary clearance, except

a. Sachharin
b. Electron microscopy
c. Scintigraphy
d. Charcoal

Ans: b